



DISCORD

Disagreement-Informed Inference for Sub-Threshold Cosmic Object Recovery and Detection

A citizen-science active-learning platform for identifying the universe's most elusive objects.

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BACKGROUND

Gravitational Microlensing

What It Is

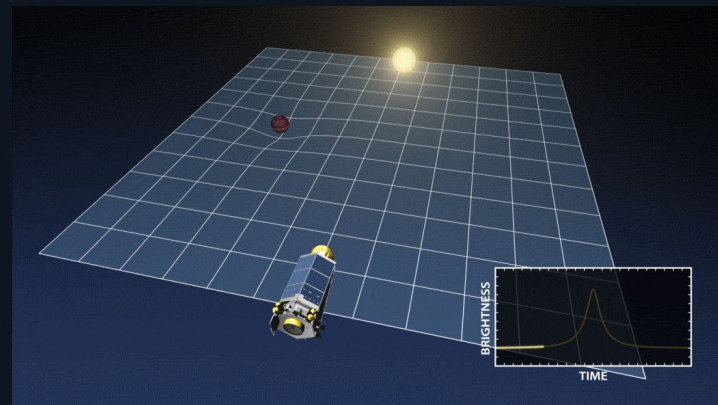
Compact objects gravitationally bend starlight, producing a distinctive flux-versus-time light curve.

Why It Matters

It is a unique method for detecting free-floating planets, primordial black holes, and dark matter subhalos.

The Problem

Exotic events often lack established templates, causing automated pipelines to overlook them.



(Alcock, 2024)

CURRENT PRACTICES

How It Is Done Today

OGLE & KMTNet

Template-fitting pipelines optimized for standard **Paczynski** light curves, rather than anomalous events.

ZTF Manual Vetting

Approximately 3,200 candidates were inspected by eye to validate only about 124 events.

Existing Citizen Science Efforts

Projects such as Planet Hunters and Astronomy typically treat disagreement as noise, averaging it away.



CURRENT PRACTICES

Current statistics

3,200

Manual Inspections

Performed in the ZTF five-year
Galactic-plane search

124

Confirmed Events

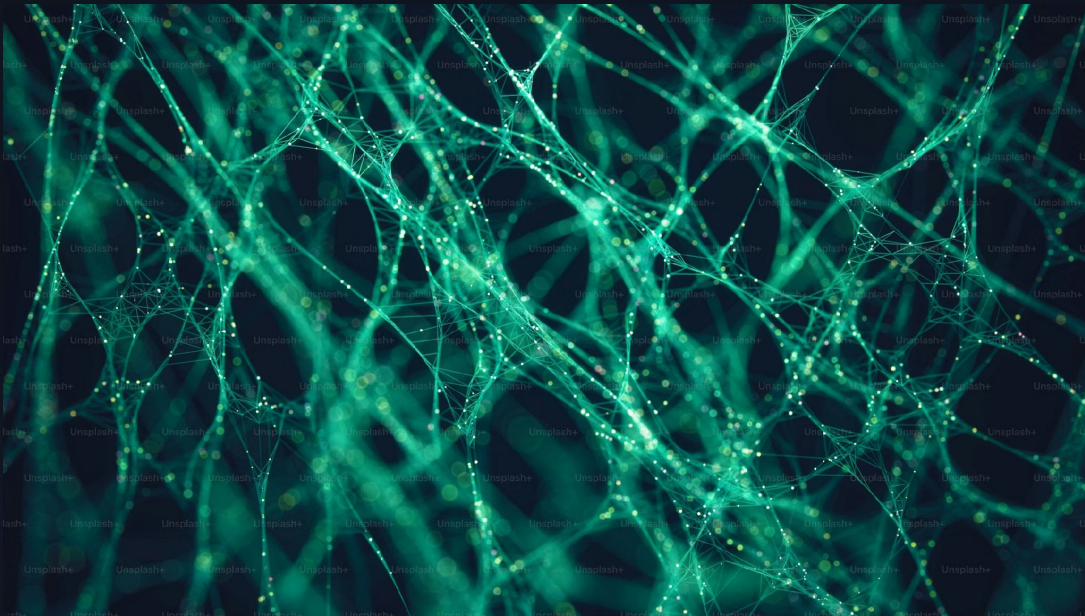
High-confidence detections
identified in that search

10M

Nightly Alerts

Rubin/LSST will begin in 2026,
exceeding the scalability of current
methods

The CNN & Active Learning Loop

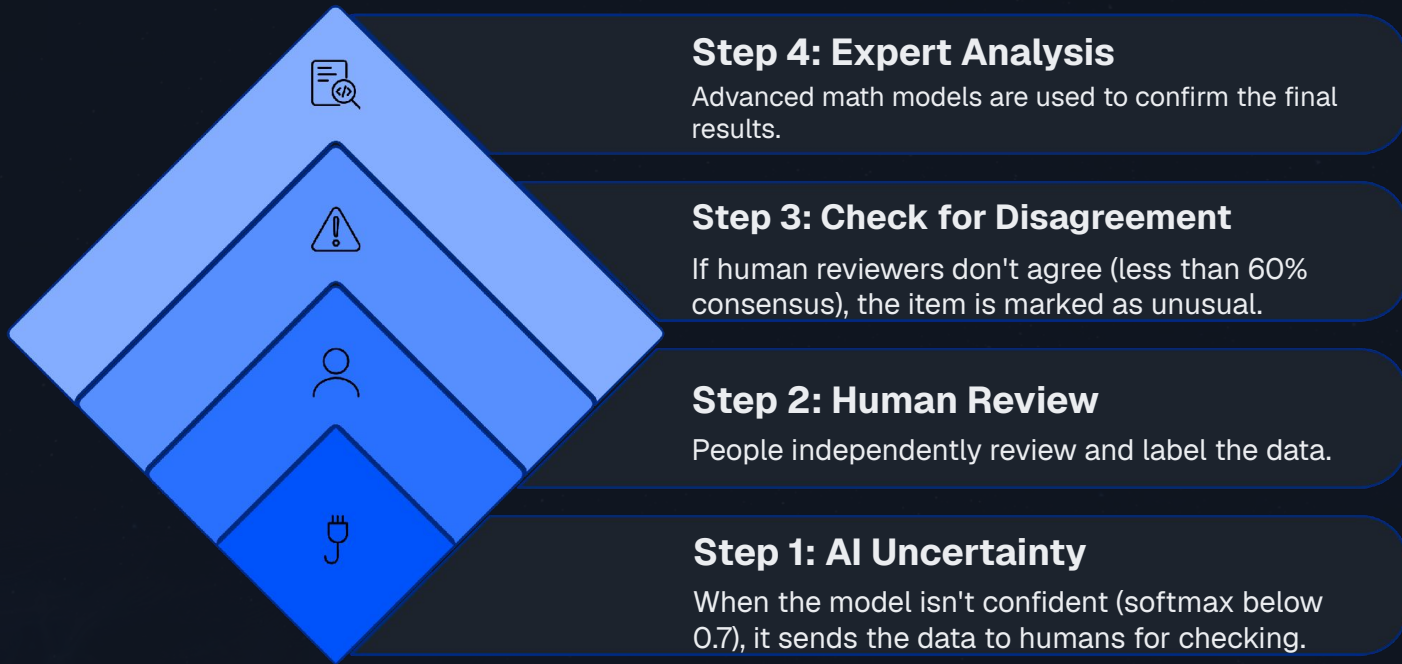


How the CNN works

CNN trained on **the disagreement distribution**, rather than the obvious majority votes that humans agreed on

- Isolate examples where human reviewers disagreed the most
- Feeds those examples into the CNN
- Idea is that these more complex and harder examples will increase how quickly the AI learns and how accurate it is at **spotting rare anomalies**

DISCORD: The architecture



This system uses two filters: AI confidence determines when humans need to help, and human disagreement determines when an expert needs to step in.

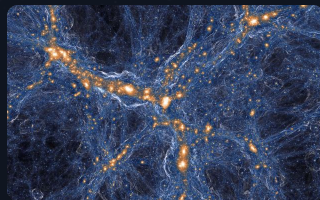
BENEFIT TO SOCIETY

Who This Serves



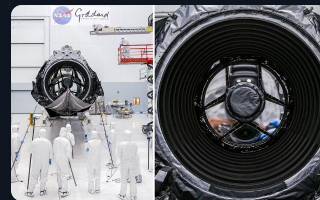
Survey Teams

OGLE, KMTNet, and ZTF benefit from reduced manual vetting burden, enabling scalable triage for LSST's 10 million alerts per night.



Cosmologists

The method identifies rare events that can constrain dark-matter substructure and primordial black hole populations.



Human-AI Synergy

An estimated 250 free-floating planets are expected; Discord-based pre-flagging can prioritize high-value candidates for follow-up observation.

Exams

Midterm

Our mid-term exam requires the baseline 1D CNN to achieve an Area Under the Curve (AUC) score of ≥ 0.85 on an explicitly defined, held-out validation set combining real light curves and simulated data (with an established positive-class prevalence)

Final

Our final exam evaluates the full active-learning loop. First, the loop must reduce the false-positive rate on background variable stars by at least 15% relative to the mid-term baseline, and reach an overall F1 score of ≥ 0.90 . Second, to directly test our research question, the disagreement-flagged pipeline must achieve a significantly higher recall rate of the synthetic injected anomalies (e.g., binary lenses, NFW-subhalos) compared to a control CNN trained on consensus labels alone.



Future Steps

Disagreement as a Signal

This work introduces disagreement-as-signal as a new framework for light-curve analysis, establishing a publishable methodological contribution.

Benefit to Broader Society

Science has always been locked away inside universities and labs. Our project changes that. Anyone with an internet connection can analyze real telescope data, spot cosmic discoveries, and help humanity understand what the universe is actually made of.

Next Steps

Our next steps for this project include expanding the pipeline for use by LSST/ZTF and trying to get as many astronomy enthusiasts and researchers using the website as possible.



**Onto the Questions
Phase**